

CMR INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Machine Learning II
Subject Code: BAI702

Assignment 1 — Machine Learning Project Based Assignment
Phase 1 Report

Disease Risk Prediction and Decision Support System:
A Probabilistic Machine Learning Approach for Early PCOS Risk Screening

Submitted by
Prakshi Lakhchaura - 1CR23AI083
Sannah Sharma - 1CR23AI110

1. Background and Justification of the Study

Polycystic Ovary Syndrome (PCOS) is one of the most common endocrine disorders affecting women of reproductive age, yet it remains one of the most inconsistently diagnosed. Global meta-analyses estimate that PCOS affects between 10% and 13% of women worldwide, with prevalence varying by diagnostic criteria: 12.1% under the Rotterdam criteria, 7.9% under the older NIH criteria, and 12.7% under the Androgen Excess Society criteria. The disease burden has nearly doubled over three decades, rising from 36.7 million affected individuals globally in 1990 to 69.5 million in 2021, with projections showing continued growth through 2036 (Global Burden of Disease Study 2021).

In India specifically, prevalence estimates range even wider, from 3.7% to 22.5% depending on the population and diagnostic framework used, with a large-scale study of over 9,800 women reporting a national prevalence of 19.6% under Rotterdam criteria. Among Indian women with PCOS, comorbidities are alarmingly common: 43.2% present with obesity, 91.9% with dyslipidaemia, 32.9% with non-alcoholic fatty liver disease, and 24.9% with metabolic syndrome. These figures establish PCOS not as a niche gynaecological concern but as a widespread metabolic and reproductive health crisis with cascading effects on cardiovascular, hepatic, and mental health.

What makes PCOS particularly suited to a machine learning intervention is the scale of its underdiagnosis. Despite affecting roughly one in ten to one in eight women, up to 70% of affected women remain undiagnosed, according to World Health Organization estimates. This gap is even more pronounced in low-resource settings, where limited access to endocrinologists, ultrasonography, and hormonal panels delays diagnosis, often until symptoms such as infertility or severe metabolic complications force clinical attention.

Even in well-resourced healthcare systems, diagnosis is complicated by the disorder's heterogeneous presentation. PCOS is not a single, uniformly expressed condition but a syndrome defined by a cluster of possible symptoms — irregular menstrual cycles, hyperandrogenism (excess facial or body hair, acne), polycystic ovarian morphology on ultrasound, insulin resistance, and weight gain — any combination of which may or may not be present in a given patient. The Rotterdam criteria, the most widely used diagnostic framework, requires only two of three core features to be present, which means two women with completely different symptom profiles can both receive the same diagnosis. This combinatorial complexity is precisely the kind of problem that rule-based or purely threshold-driven clinical diagnosis struggles with, but that probabilistic machine learning models are well equipped to handle, since they can learn the relative weight and interaction of each contributing factor directly from patient data rather than relying on a fixed checklist.

There is also a strong justification rooted in health equity. PCOS disproportionately affects women during their most economically and socially active years, typically between ages 15 and 44, and its consequences extend well beyond fertility. Untreated or late-diagnosed PCOS is associated with elevated long-term risk of type 2 diabetes, cardiovascular disease, endometrial cancer, and significant psychological burden including anxiety and depression linked to symptoms like hirsutism, acne, and weight gain. Because early symptoms are frequently dismissed as “normal” period irregularity or attributed to lifestyle factors, women often see multiple healthcare providers over several years before receiving a diagnosis. A decision-support tool that can flag elevated PCOS risk from readily available clinical and lifestyle parameters, before a full diagnostic workup is pursued, could meaningfully shorten this diagnostic delay, particularly for women in primary care or low-resource settings who may not have immediate access to specialist care.

From a technical standpoint, the domain is also well matched to available machine learning resources. Public datasets such as the Kaggle PCOS dataset (compiled by Prasoon Kottarathil from 10 hospitals in Kerala, covering 44 clinical, biochemical, and ultrasound features across 541 patients) provide a credible foundation for model development within an academic timeline. Prior studies using this and comparable datasets have demonstrated that models ranging from Naive Bayes to Random Forest and ensemble methods can achieve accuracies from 81% to over 96%, confirming that the signal-to-noise ratio in PCOS clinical data is favourable for supervised learning. This existing body of work provides both a benchmark to build against and a clear gap to address: most published PCOS prediction studies optimize purely for classification accuracy and treat the model as a black box, with comparatively little emphasis on probabilistic risk quantification, interpretability of which parameters drive a given prediction, or integration into a usable decision-support framework a clinician could actually act on.

This project is therefore justified on three converging grounds. First, PCOS represents a large and growing public health burden that is substantially underdiagnosed relative to its true prevalence, particularly in India, where this project's data and context are most directly relevant. Second, the syndrome's multifactorial, criteria-based diagnosis is inherently probabilistic in nature, making it a strong conceptual fit for probabilistic machine learning models rather than a rigid rule-based system. Third, sufficient real-world clinical data already exists in accessible form to make a rigorous, evidence-based model achievable within the scope of an academic project, while leaving clear room for contribution through improved interpretability and decision-support design rather than accuracy alone. Together, these factors motivate a project that is not only technically tractable but addresses a genuine and underserved gap in women's healthcare.

Sources: *WHO PCOS Fact Sheet* (who.int); *Prevalence of PCOS — Global Meta-Analysis* (academic.oup.com); *Global Burden of PCOS 1990–2021* (pmc.ncbi.nlm.nih.gov); *PCOS Prevalence in India — Systematic Review* (pmc.ncbi.nlm.nih.gov); *Prevalence, Phenotypes, Comorbidities among Indian Women* (pmc.ncbi.nlm.nih.gov); *Kaggle PCOS Dataset* (kaggle.com).

2. Goals and Scope of the Project

Overall Goal

- Design and evaluate a probabilistic machine learning-based decision support system that predicts a woman's likelihood of having PCOS using routinely collected clinical, biochemical, and lifestyle parameters.
- Output an interpretable, clinically actionable risk estimate rather than a simple binary label.
- Function as a triage/screening aid, flagging elevated risk early using inexpensive or already-available data, so women can be prioritized for confirmatory clinical evaluation sooner. It is explicitly not intended to replace clinical diagnosis, which requires ultrasonography, clinical judgment, and exclusion of related endocrine disorders under Rotterdam/NIH criteria.

Specific Objectives

- **Data sourcing** — Consolidate a reliable patient health dataset covering clinical, biochemical, and ultrasound-derived PCOS features, primarily the Kaggle PCOS

dataset (Kerala hospitals, 541 patients, 44 features), supplemented where feasible by UCI or IEEE DataPort sources.

- **Preprocessing** — Handle missing values, encode categorical variables, normalize biochemical measurements on different scales (e.g., LH/FSH ratio vs. BMI vs. follicle count), and correct class imbalance (roughly one-third PCOS-positive cases in the primary dataset).
- **Model building** — Build and compare probabilistic classifiers (Naive Bayes, Bayesian Networks) against non-probabilistic baselines (Logistic Regression, Random Forest) to test whether probabilistic framing adds value beyond accuracy, particularly for expressing predictive uncertainty.
- **Feature analysis** — Analyse the relative influence of individual parameters (follicle count, BMI, cycle irregularity, hormonal ratios) on predictions, so the system explains its output rather than acting as a black box.
- **Evaluation and prototyping** — Evaluate models with standard classification metrics and prototype how output could be surfaced through a decision-support interface for a healthcare provider or patient self-screening.

In Scope

- Publicly available, de-identified datasets only — no primary data collection (which would require ethical clearance and clinical partnerships beyond a coursework project).
- Structured, tabular clinical data — including numeric features derived from ultrasound reports (e.g., follicle count, ovarian volume).
- At least two trained probabilistic models with documented preprocessing pipelines and hyperparameters.
- Evaluation using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis — chosen because false negatives (missed PCOS cases) and false positives (unnecessary alarm) carry different real-world costs, making accuracy alone insufficient.
- Interpretability analysis via feature importance rankings or SHAP values.
- A written discussion of how model output could support, not replace, clinical decision-making.

Out of Scope

- Definitive diagnosis, treatment prescription, or clinical/regulatory deployment — this remains an academic research prototype.
- Image-based diagnosis (e.g., CNN classification of raw ultrasound images) — a separate, substantial research area.
- Other women's health conditions beyond PCOS in this phase, though the framework is designed to be conceptually extensible (e.g., to endometriosis or gestational diabetes) in future work.
- A full-scale user study with patients or clinicians, given time constraints; usability/adoption considerations are discussed qualitatively instead.

Boundaries and Caveats

- The primary dataset originates from Kerala, India — conclusions about model performance are most directly applicable to that population, with generalizability to other populations flagged as a limitation, not assumed.
- The project produces a validated model and a functional prototype/mock-up of a decision-support interface, not a production-grade or regulatory-compliant medical device.

3. Role of Machine Learning in Addressing the Problem

Why Traditional Approaches Fall Short

- PCOS diagnosis relies on criteria (Rotterdam: 2 of 3 features present) that were designed for manual clinical judgment, not systematic risk scoring — this leaves room for inconsistent application across clinicians and missed borderline cases.
- Symptom presentation is highly heterogeneous — two patients with entirely different symptom combinations can both meet criteria, making fixed rule-based checklists a poor fit for capturing the full risk gradient.
- Manual screening depends on specialist access (endocrinologists, sonographers), which is scarce in low-resource and rural settings — a key driver of the estimated 70% underdiagnosis rate.
- Traditional statistical approaches struggle to capture non-linear interactions between features, such as how BMI and insulin resistance jointly amplify hormonal imbalance risk.

Core Contributions of Machine Learning to This Problem

- **Pattern recognition across many correlated features** — ML models can simultaneously weigh 15–40+ clinical, biochemical, and lifestyle variables and learn which combinations are most predictive, something manual scoring cannot practically do.
- **Probabilistic risk quantification** — rather than a binary output, probabilistic models (Naive Bayes, Bayesian Networks) produce a likelihood score, better reflecting the genuine diagnostic uncertainty inherent in a syndrome defined by partial-criteria matching.
- **Uncertainty-aware reasoning** — Bayesian approaches allow prior clinical knowledge to be combined with patient-specific evidence, updating risk estimates as more data becomes available — mirroring how a clinician reasons, but systematically.
- **Non-invasive, low-cost screening** — models trained on easily collected data can flag risk before expensive tests are ordered, reducing cost and diagnostic delay.
- **Feature importance and interpretability** — techniques like SHAP values reveal which parameters drive a given prediction, critical for clinician trust and adoption.
- **Handling missing and noisy real-world data** — ML preprocessing pipelines and models robust to missing values make it feasible to work with imperfect real-world patient records.
- **Scalability** — once trained, a model can screen large populations or be embedded in telemedicine/self-assessment tools at near-zero marginal cost.

Specific Techniques and Their Roles

- **Naive Bayes / Bayesian Networks** — primary probabilistic models for this project; well suited to expressing risk as a probability and reasoning under uncertainty with correlated symptom features.
- **Logistic Regression** — baseline probabilistic-adjacent model, useful for interpretable coefficient-based comparison against Bayesian approaches.
- **Random Forest / Ensemble methods** — benchmark for raw predictive accuracy (prior PCOS studies report 85–96%+ accuracy).
- **SHAP / feature importance analysis** — layered on top of the above models to analyze the influence of individual health parameters on prediction.

Role in Decision Support, Not Diagnosis

- ML's role here is explicitly to augment, not replace, clinical judgment — output is a risk flag and explanatory feature breakdown, intended to prompt earlier referral or testing.
- This directly addresses the underdiagnosis problem identified in Section 1 by shortening the path from symptom onset to clinical evaluation.
- It also supports consistency — a validated model applies the same learned criteria to every patient, reducing variability in early-stage screening decisions.

Machine learning is used here not simply because the problem involves data, but because PCOS diagnosis is fundamentally a probabilistic, multi-factorial pattern-matching problem operating on incomplete, heterogeneous clinical data — exactly the class of problem where probabilistic ML models outperform fixed rule-based or purely manual approaches, while remaining interpretable enough to fit into a real clinical decision-making workflow.

4. Summary of Related Research and Prior Approaches

Foundational Studies on the Kaggle PCOS Dataset

- Multiple studies using the Kaggle PCOS dataset (Kottarathil, 541 patients, 44 features from 10 Kerala hospitals) established it as the standard benchmark for this problem space.
- Aggarwal et al. (2023) applied statistical feature selection combined with a Random Forest classifier, achieving 93.52% accuracy — demonstrating that feature selection meaningfully improves performance over using all 44 raw features.
- A separate evaluation found Random Forest, Multilayer Perceptron, and Linear SVM to be the most performant algorithms overall, with accuracy ranging from 85% to 93%.
- Some later studies report even higher accuracy (99.31–99.32%) using aggressive feature selection and 8-fold cross-validation, though such high figures on a relatively small (541-patient), single-region dataset raise legitimate concerns about overfitting rather than genuine generalizable performance.

Probabilistic and Bayesian Approaches

- A study comparing Decision Tree and Naive Bayes models on 100 PCOS-positive and 100 control cases found Naive Bayes achieved more consistent performance: 81% test accuracy and an F1-score of 0.81, concluding it shows promise as a simple, non-

invasive screening tool, especially for primary care and low-resource settings (Indian Journal of Community Health).

- This directly supports the present project's choice of probabilistic models: even though Naive Bayes underperforms ensemble methods on raw accuracy, its calibrated probability outputs and interpretability make it more suitable for a decision-support context than a black-box high-accuracy classifier.
- Broader literature on Bayesian Networks in medical diagnosis confirms their established role in modelling dependencies and uncertainty between clinical variables, reinforcing the methodological basis for extending Naive Bayes to a fuller Bayesian Network in this project.

Explainable AI and Interpretability-Focused Studies

- SmartScanPCOS (Heliyon, 2024) combined a two-level Random Forest classifier with SHAP-based explainability, aiming to make feature contributions (hormone levels, BMI, follicle count) transparent to clinicians rather than only optimizing accuracy.
- The EAIBS-PCOS system (2025) combined explainable AI with blockchain-based data handling, reporting 98% accuracy, reflecting a growing trend of pairing high performance with interpretability and data governance.
- A non-invasive PCOS diagnosis study (Scientific Reports, 2025) used chi-square-based SelectKBest feature selection validated against XGBoost feature importance and SHAP analysis, identifying the top 10 predictive features — a methodological approach directly relevant to this project's own feature-influence analysis objective.
- A 2025 systematic review following PRISMA 2020 guidelines catalogued AI applications in PCOS across diagnostic performance, biomarker discovery, risk prediction, decision support, and interpretability, indicating the field is actively consolidating around explainability as a research priority, not just accuracy.

Broader Comparative and EHR-Based Studies

- A study evaluating Random Forest, Gradient-Boosted Trees, SVM, and Logistic Regression on electronic health record data reported AUCs between 77.4% and 85%, with Gradient-Boosted Trees performing best (Frontiers in Endocrinology, 2024) — notably lower than Kaggle-dataset studies, suggesting real-world EHR data is noisier and harder to model than curated research datasets.
- A stacked learning framework study addressed class imbalance and feature selection jointly, reporting improved robustness over single-model baselines, highlighting imbalance handling as a recurring technical concern also flagged in this project's scope.
- A separate study explored radial pulse wave analysis as a novel, fully non-invasive feature source for PCOS classification, illustrating an active research trend toward reducing reliance on invasive lab testing.

Identified Gaps This Project Addresses

- **Generalizability** — most high-accuracy results come from small, single-region, single-hospital-system datasets; performance drops noticeably on more heterogeneous EHR data, a limitation this project will explicitly acknowledge rather than obscure behind a single accuracy figure.

- **Accuracy-only framing** — many studies report a single top accuracy number without adequately addressing class imbalance costs; this project's evaluation plan directly addresses this via precision/recall/F1/ROC-AUC/confusion matrix analysis.
- **Interpretability as an add-on rather than core design** — most earlier Kaggle-dataset studies treat models as black boxes; this project builds interpretability into the core objectives from the outset via inherently probabilistic models.
- **Decision-support framing** — very few studies translate model output into an explicit discussion of clinical workflow integration; this represents a genuine, achievable contribution area for this project.

Sources: Springer — ML Techniques on Kaggle PCOS Dataset; Indian Journal of Community Health — Probabilistic Screening Tool for PCOS; MDPI — Bayesian Networks Scoping Review; Heliyon — SmartScanPCOS; PeerJ — EAIBS-PCOS; Scientific Reports — Non-invasive PCOS Diagnosis; Frontiers in Endocrinology — EHR-based PCOS Prediction; PMC — Stacked Learning Framework; PMC — AI Systematic Review in PCOS.

5. Planned Framework / Proposed Solution Design

High-Level Architecture

- **Layer 1 — Data Layer:** raw patient health data (Kaggle PCOS dataset as primary source, supplemented by UCI/IEEE DataPort if needed) → cleaned, structured dataset.
- **Layer 2 — Processing Layer:** preprocessing, feature engineering, and feature selection pipeline.
- **Layer 3 — Modelling Layer:** probabilistic classifiers (Naive Bayes, Bayesian Network) trained alongside baseline models (Logistic Regression, Random Forest) for comparison.
- **Layer 4 — Evaluation Layer:** performance metrics, interpretability analysis (feature importance/SHAP), calibration checks on predicted probabilities.
- **Layer 5 — Decision Support Layer:** a prototype interface (web mock-up or notebook-based dashboard) that takes patient inputs and outputs a risk probability with an explanation of contributing factors.

Step 1: Data Collection

- Primary dataset: Kaggle PCOS dataset (541 patients, 44 clinical/biochemical/ultrasound features, binary PCOS label).
- Supplementary sources considered: UCI Machine Learning Repository and IEEE DataPort PCOS data, to assess feasibility of combining datasets for improved generalizability (flagged as a stretch goal given differing feature sets across sources).
- Data documentation: record source, collection period, population characteristics, and known limitations for transparency in the final report.

Step 2: Data Cleaning and Preprocessing

- **Missing value handling:** assess missingness pattern per feature; apply mean/median imputation for continuous variables and mode imputation for categorical variables, with sensitivity checks.

- **Outlier detection:** use IQR or z-score methods on continuous biochemical variables (LH, FSH, AMH, TSH) to flag implausible lab values.
- **Feature scaling:** normalize/standardize continuous features with widely different ranges so no single feature dominates purely due to scale.
- **Categorical encoding:** one-hot or ordinal encoding for categorical symptom indicators (e.g., hirsutism, acne, hair loss).
- **Class imbalance correction:** apply SMOTE or class-weighted loss functions to address the roughly 2:1 imbalance between PCOS-negative and PCOS-positive cases.
- **Train/validation/test split:** stratified split (e.g., 70/15/15) with k-fold cross-validation (5- or 8-fold) used during model selection.

Step 3: Feature Analysis and Selection

- Correlation analysis to identify multicollinearity among the 44 raw features.
- Statistical feature selection (chi-square, ANOVA F-test, or SelectKBest) to rank features by predictive relevance.
- Cross-validation of statistically selected features against model-derived importance (e.g., Random Forest importances, SHAP values) to identify a consensus feature subset.

Step 4: Model Development

- **Naive Bayes** (Gaussian for continuous features) as the baseline probabilistic model, valued for simplicity, speed, and calibrated probability outputs.
- **Bayesian Network**, built via expert-informed structure or structure-learning algorithms, to capture conditional dependencies Naive Bayes' independence assumption ignores.
- **Baseline comparison models:** Logistic Regression, Random Forest, and optionally SVM, to contextualize probabilistic model performance against established high-accuracy approaches.
- **Hyperparameter tuning:** grid search or randomized search with cross-validation for each model, documented for reproducibility.
- **Tooling:** Python with scikit-learn, pgmpy (Bayesian Networks), pandas/NumPy, and matplotlib/seaborn for visualization.

Step 5: Evaluation Plan

- Standard classification metrics: accuracy, precision, recall, F1-score, and ROC-AUC, computed on the held-out test set.
- Confusion matrix analysis to explicitly examine the false negative rate, given its higher clinical cost relative to false positives in a screening context.
- Probability calibration assessment (calibration curves, Brier score) specifically for the probabilistic models.
- Comparative results table across all models.

Step 6: Interpretability and Decision Support Prototype

- Generate SHAP summary plots (or feature-importance bar charts) showing which features most influence predictions overall and for individual patient cases.

- Build a simple prototype interface (e.g., a Streamlit web app) where a user inputs patient parameters and receives a risk probability, a plain-language summary of top contributing factors, and a recommendation to seek clinical confirmation if risk exceeds a defined threshold.
- Design the output explicitly as a screening aid with appropriate disclaimers, not a diagnostic verdict.

Validation and Robustness Checks

- Sensitivity analysis: re-run key models with different imputation strategies and feature subsets to check result stability.
- Compare probabilistic model performance against literature benchmarks to sanity-check results and investigate any large divergence.

6. Anticipated Results and Contributions

Expected Model Performance

- Based on comparable studies using the same Kaggle PCOS dataset, baseline models are expected to achieve accuracy in the 80–93% range: Naive Bayes around 80–85%, Bayesian Network somewhat higher due to modelling feature dependencies, and Random Forest/Logistic Regression baselines in the upper end (90%+).
- Probabilistic models are anticipated to show a modest accuracy trade-off relative to ensemble methods but are expected to produce better-calibrated probability estimates (lower Brier score, more reliable calibration curves), which matters more than raw accuracy for a risk-communication tool.
- Precision/recall are expected to reveal a meaningful trade-off between catching more true PCOS cases and reducing false alarms; this will be explicitly analysed rather than resolved by a single “best” threshold.
- Feature importance analysis is expected to confirm known clinical risk factors as top predictors (BMI, follicle count, cycle irregularity, LH/FSH ratio), serving as a validity check that the model has learned clinically meaningful patterns.

Anticipated Technical Contributions

- A documented, reproducible preprocessing pipeline addressing missing data, class imbalance, and feature scaling specifically tuned for the PCOS dataset's known quirks.
- A direct empirical comparison between probabilistic models and standard high-accuracy baselines on the same data split, explicitly reporting calibration quality alongside accuracy.
- A consolidated feature-importance analysis cross-validated across statistical selection methods and model-derived importance scores.
- A working prototype decision-support interface translating model output into a probability plus plain-language explanation.

Anticipated Contributions to the Problem Domain

- Evidence on whether probabilistic models offer genuine practical advantages that justify their modest accuracy trade-off relative to black-box ensemble methods, directly relevant to real-world adoption in low-resource clinical settings.
- A clearer articulation of which health parameters most reliably signal PCOS risk, informing lightweight, low-cost screening questionnaires that do not require full biochemical panels.
- A demonstration, even at prototype scale, of how a decision-support tool could shorten the diagnostic delay associated with PCOS.

Benefits for Healthcare Professionals

- A secondary screening signal for primary-care physicians without ready access to endocrinology referral, helping prioritize which patients need urgent specialist follow-up.
- Reduced reliance on clinical intuition alone for borderline cases where Rotterdam criteria are only partially met.
- Interpretable feature explanations allow clinicians to sanity-check the model's reasoning against their own clinical judgment.
- Potential reduction in unnecessary invasive or costly testing for patients confidently flagged as low risk, though this would require clinical validation beyond this project's scope.

Benefits for Patients

- Earlier awareness of PCOS risk, potentially shortening the multi-year diagnostic delay reported in prior research.
- A low-cost, non-invasive first step that does not require an immediate clinic visit or expensive lab work, lowering the barrier to seeking care.
- Greater clarity on personal risk factors that could motivate earlier lifestyle intervention, itself a first-line PCOS management strategy.

Limitations to Be Transparently Reported

- Results will be specific to the training population (Kerala, India) and should not be assumed to generalize globally without further validation.
- As a prototype, the system will not undergo clinical trial validation; anticipated benefits are presented as plausible outcomes based on the model's design and prior literature, not as proven clinical impact.
- Dataset size (541 patients) limits the complexity of Bayesian Network structures that can be reliably learned without overfitting.